

How Informed Are Foreign Institutional Investors in India?

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Abstract:

How informed are Foreign Institutional Investors (FIIs) in Indian financial markets? We build a model of imperfect competition in a financial market, in which market makers have up to two public signals on which to define their pricing rule – firms' earnings announcements and FII trading signals. We define the payoff structure with a sum of correlated components, one known to firms, and the other, to FIIs. We estimate the deep parameters of the model such as the variance governing the FIIs' informational advantage, the level of background noise, and the correlation between the two components of the payoff. So instead of treating multicollinearity as a problem to be resolved using more orthogonal instruments, or a better selection of regressors, we explain it using an underlying model of financial market equilibrium, and identify deep parameters that are unobservable but have economic significance.

Our results indicate the information advantage of FIIs with respect to the component they have information about, exceeds the information advantage that firms have with respect to information released via earnings announcements. We also find that correlation between the two fundamental information components is significantly positive, so that in econometric work we should consider such an environment, even beyond any correlation arising from imperfect measurement. A methodological contribution is to show how the underlying model of equilibrium allows us to learn more from the event study based on earnings announcements.

Keywords: Foreign Institutional Investors, Institutional Trading, Earnings Announcements, Investment Performance.

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1. Introduction

Foreign institutional investors (FIIs) have become prominent in emerging markets. But one elementary question has not been resolved: to what degree are FIIs informed traders who use a careful optimizing strategy, and to what degree are they, like many other institutional traders, participating in the market primarily because of retail pressures to meet insurance or pension claims or to respond to portfolio rebalancing decisions made by their customers with needs of their own? This is the main empirical question that we address in this paper.

We pose the question within the context of a model of asset pricing under imperfect competition that not only allows for variation in FII type, but one in which a key source of public information, earnings, is also sometimes available. Therefore, the competitive price-setters in our model observe either or both of two signals, corporate earnings announcements and FII trading signals, before setting prices. While FII trading signals are observable, whether or not they reflect information and strategy more, is not. That noise helps preserve FII incentives to gather potentially costly private information.

We specify firm payoff information as a sum of component information innovations, *a la* Admati and Pfleiderer (1988), one component known to the firm, and another, to FIIs. But we allow these components to be correlated, which creates a rich enough environment to capture a variety of relationships between the two signals. But the model is also parsimonious enough to allow estimation of even the deep parameters of the model, such as the variability of FIIs' private information, noise, and the correlation between corporate earnings public signal and FIIs' private information component. Estimates of the deep parameters, in turn, let us quantify

various aspects of FII behavior that so far have only been discussed speculatively, and opens up additional questions.

Our study employs a database of daily stock-level trades of FIIs in India for the years 2003-2015. We integrate data on quarterly earnings announcements and stock returns from the PROWESS database with FII trades during the announcements to conduct tests of FII informedness. We find that the information advantage of FIIs with respect to the component they have information about, exceeds the information advantage that firms have with respect to information released via earnings announcements. We also find that correlation between the two fundamental information components is significantly positive, so that in econometric work we should consider such an environment, even beyond any correlation arising from imperfect measurement.

The rest of the paper is organized as follows. In section 2, we provide a brief review of prior literature that relates to our research. Section 3 develops the model, and describes equilibrium properties. Section 4 describes our data sources and sample selection process and section 5 presents the empirical variable definitions and main empirical models. In section 6, we present our results, and in section 7 we discuss our conclusions.

2. Prior Literature

Admati (1985) generalized the single-security noisy rational expectations model under perfect competition due to Hellwig (1980) to the case of multiple securities, allowing for general variance-covariance matrices governing payoffs, errors in private signals and liquidity noise. She showed that a common intuition in a single-security setting, that a security price would be increasing in its own payoff, need not hold with many securities and sufficient correlation. Caballé and Krishnan (1994) generalized the risk-neutral imperfect competition model due to Kyle (1985), to the case of N assets and K traders, with a similarly

rich correlation structure, and showed that asset prices again need not be increasing in their own payoffs. They also showed that portfolio diversification can arise for reasons unrelated to risk, to minimize revelation of information in a correlated environment.

Lundholm (1988), under perfect competition, and Manzano (1999), under imperfect competition, showed that even with one security, if there were multiple signals available, for example, a public signal like earnings together with private signals for each trader, a similar ambiguity can arise. A security's price may not rise in earnings. The key to this result is the information structure used in both these papers, where an asset has payoff v and the signals, public and private, are of the form $s_i = v + e_i$, with $\text{Cov}(e_i, e_j)$, $i \neq j$, not necessarily zero. In this case, each signal has both a direct and an indirect effect. A large value of s_i could indicate a high v , this is the direct effect. On the other hand, it could indicate a large e_i , and, if the covariance between errors in signals is high enough, also a high e_j , $j \neq i$, and consequently a lower v ; this is the indirect effect. When the indirect effect dominates, good news can be bad news.

In our work here, the payoff is defined with additive components, and corporate earnings and traders' private information are about different components. The correlation parameter can lead to an ambiguous coefficient on earnings in the pricing rule, but the exact reason is a little different, as in our case the correlation parameter governs two different payoff components, and not the errors on the same component in two different signals.

3. Model

Our primary model is a model of asset pricing under imperfect competition with both public and private signals. The main objective is to ensure that it is rich enough to address the empirical question about the degree to which FIIs are informed and strategic, while being simple enough to admit of easy estimation of even primitive parameters such as the correlation between public and private information, the informational advantage of FIIs, and the level of noise.

3.1. Assumptions

(A1) Assets, asset payoffs, and information about asset payoffs: There is one risky asset, and one riskless asset (numeraire) with payoff and price equal to one. The payoff to the risky asset (and equivalently, information about this payoff) is given by $\tilde{v}$, $\tilde{v} = \tilde{v}_F + \tilde{v}_T$, where $\tilde{v}_F$ is the informational innovation component for which the firm has an informational advantage relative to others (captured in the data by unexpected earnings); and $\tilde{v}_T$ is the informational innovation component for which the institutional traders (FIIs) have an informational advantage relative to others. The components $\tilde{v}_i \sim N(0, \sigma_i^2)$, $i = F, T$, with $\text{Cov}(\tilde{v}_F, \tilde{v}_T) = \rho \cdot \sigma_F \cdot \sigma_T$, $\rho \in (-1, 1)$, $\sigma_i > 0$, $i = F, T$.

This component structure for payoffs has been used before in, e.g. Admati and Pfleiderer (1988). But where they chose to make the total payoff a sum of orthogonal components, we allow the two components to be correlated. This is important in creating a setting where the two signals that price-setters observe – earnings announcements and trading signals – can be substitutes, complements or independent. This allows for more possibilities than with the more common structure where both public and private signals are about the same component, yet retains parsimony in terms of the number of parameters to be estimated.

This structure provides additional advantages. We could interpret $\tilde{v}_i$ as perfect information on a component observable to $i, i = F, T$, and for ease of exposition we will sometimes do that. But occasionally it will be more convenient to interpret it as the informational advantage of i relative to others, i.e. $\tilde{v}_i = E(v|I_i) - E(v)$. By not having to specify the signals in the information sets I_i we can save on some additional parameters while being slightly more general.¹ It does mean that the variance parameters σ_F^2 and σ_T^2 serve both as prior variances and as measures of informational advantage. The variance parameter σ_F^2 governing earnings can be estimated directly from the data, and so provides a convenient scale variable with which to interpret the magnitude of any estimates of σ_T^2 .

It provides a simple way of describing whether firms know more or less than traders, without adding more notational burden. We may think of firms as having an advantage with respect to events within the firm, captured in the judgment reported in its summary earnings number; and of traders as having an advantage with respect to events outside the firm, which may include analysis of the competition, the links in the supply chain, government policy and even the state of the financial market. But this interpretation is only a suggestion. It is not essential.

(A2) Agents:

(i) Firm: There is a firm, denoted by subscript $i = F$, which observes $\tilde{v}_F = v_F$ perfectly and reports it faithfully, as required to do so under accounting rules. Note, however, that because of the component structure of total firm payoff, seeing and reporting perfect information on one component is not same as knowing and reporting “everything.” Our assumption A1 above allows firms to know a lot or little. One interpretation is that auditing works and results in compliance (see, e.g. Shin (1994)). Alternatively, we could invoke models of cheap talk (for example, Bhattacharya and Krishnan (1999)) in which firms have

¹ For a convenient summary of the algebra of informational advantages, see the remarks following assumption (A3) in Caballe and Krishnan (1994).

an incentive to make truthful disclosures, despite being able to lie with impunity. In either case, this assumption is broadly consistent with the vast empirical literature that has documented a consistent positive association between unexpected earnings and abnormal returns, while also noting that only a small portion of price variation is explained by earnings variation, even within an earnings announcement window.

(ii) Noise trader: This trader generates a random net demand of $\tilde{z}$, $\tilde{z} \sim N(\mathbf{0}, \sigma_z^2)$, $\tilde{z}$ uncorrelated with payoff components. This is intended to capture the non-strategic or non-information-based activity of FIIs.²

(iii) Strategic trader: This trader, denoted by subscript $i = T$, is intended to capture the behavior of strategic informed FIIs. She chooses a demand for the risky security, x , based on all information available to her: the public signal created by the firm's earnings announcement, v_F , and the perfect private signal about the second component, $\tilde{v}_T = v_T$, and the noise trade of some FIIs, z . Being able to observe z is different from Kyle (1985), but similar to Rochet and Vila (1994). Therefore, the strategic trader is not just better informed than the market makers (who can only observe the aggregate FII demand, $\omega = x + z$), but their information is nested in hers.³

(iv) Competitive market makers: We assume that there are competitive risk-neutral market makers whose competition makes them earn zero expected profits, so the price they set for the risky asset is equal to the expected payoff from the security given all publicly available information. In the main model of this paper, that public information will consist of the earnings signal that the firm provides, v_F , and the aggregate FII order flow, $\omega = x + z$.

Hence, the price $p = E(v|v_F, \omega)$. We also assume a linear pricing rule, $p = \alpha + \beta v_F + \lambda \omega$.

² We have studied a variant of the model in the paper that allows for z to be correlated with a payoff component. It is possible to compute equilibrium even in such a model, but the added analytical complexity yields no additional intuition about FII or market behavior, but complicates parameter estimation substantially.

³ Rochet and Vila (1994) adopt this for an important theoretical reason. Given a nested information structure, and exogenous total profits in the game, they show uniqueness of equilibrium under otherwise very general assumptions. In the Kyle (1985) framework, uniqueness has only been shown given a linear pricing rule, and uniqueness of equilibrium in general is still an open question.

Given the uniqueness of equilibrium result in Rochet and Vila (1994), this ex ante assumption of a linear pricing rule is not really an additional restriction, but makes the solution procedure more convenient.

Notice that while the aggregate FII trading signal is observable (as assumed in our empirical work), because some FIIs may trade for non-informational reasons, while other FIIs are informed and strategic, the FII private information, v_T , cannot be unraveled. Unlike Rochet and Vila (1994) we make this assumption here primarily for empirical reasons. In the empirical sequel later in this paper, we will associate the FII trading signal with the aggregate demand from all kinds of FIIs, i.e. with $\omega = x + z$. In a paper where the primary focus is on FII behavior we implicitly regard other kinds of strategic and noise traders as being of at best second-order importance and so we ignore their behavior.

3.2. Definition of equilibrium

An equilibrium of this model is defined by a trader (FII) strategy $x(v_F, v_T, z)$ and a pricing rule $p = \alpha + \beta v_F + \lambda \omega$, such that we have

- (i) Trader optimization: Given the above pricing rule, and any triple of realized values $\{v_F, v_T, z\}$ the trader T has a demand strategy $x(v_F, v_T, z)$ that is at least as good as any alternate strategy $x'(v_F, v_T, z)$.
- (ii) Market efficiency: for any realization of earnings v_F and aggregate FII order flow $\omega = x + z$, the price $p = E(v|v_F, \omega)$.

3.3. Properties of equilibrium

Proposition 1: The unique equilibrium of this model is defined by

- (i) a trader (FII) strategy $x(v_F, v_T, z) = \tau_0 + \tau_1 v_F + \tau_2 v_T + \tau_3 z$, where

$$\tau_0 = 0, \tau_1 = \left(\frac{-\rho\sigma_z}{2\sigma_F(\sqrt{1-\rho^2})} \right), \tau_2 = \left(\frac{\sigma_z}{2\sigma_T(\sqrt{1-\rho^2})} \right), \tau_3 = -\left(\frac{1}{2} \right), \text{ and}$$

(ii) a pricing rule $p = \alpha + \beta v_F + \lambda \omega$, where

$$\alpha = 0, \beta = 1 + \rho \left(\frac{\sigma_T}{\sigma_F} \right), \lambda = \left(\frac{\sigma_T(\sqrt{1-\rho^2})}{\sigma_Z} \right).$$

The proof is outlined in the appendix. In the key final step, we equate coefficients in the pricing rule, to get three equations of the form, $\alpha = f_1(\alpha, \beta, \lambda)$, $\beta = f_2(\alpha, \beta, \lambda)$, $\lambda = f_3(\alpha, \beta, \lambda)$. From the first alone, it is easy to show that $\alpha = 0$. Manipulating the other two leads to a cubic in two variables, β and λ , instead of in λ alone as in Kyle (1985) and Rochet and Vila (1994). Of the three solutions, only one satisfies $\lambda > 0$, which is needed to satisfy second-order conditions. So we have a unique real root. The solution is easily verified.

The intercepts being zero only reflects prior zero means of all variables. The coefficient τ_1 , which represents the weight the trader places on v_F , shows the effect of the correlation between the two payoff components. Though v_F provides perfect information about one component and is public, the expression for τ_1 is complex because it also yields information about the second component, as $E(v_T|v_F) = \rho \left(\frac{\sigma_T}{\sigma_F} \right) v_F$, so that given v_F the trader effectively faces both a different intercept and slope. The coefficient τ_1 is increasing in the ratio $\left(\frac{\sigma_Z}{\sigma_F} \right)$ for any $\rho > 0$; decreasing in that ratio, for any $\rho < 0$. It is also decreasing in ρ .

The parameter β can even be negative. But the reason for a possible counter-intuitive sign (causing good news to be bad news) is different in this setting from the reason in Lundholm (1988) and Manzano (1999). In those papers the multiple signals are all about the same component and correlation governs the error covariance, so β (coefficient of the public signal in the price function) in those models can be negative when the indirect effect dominates the direct effect for sufficiently large positive error covariance. Here $\beta < 0 \Leftrightarrow \rho \left(\frac{\sigma_T}{\sigma_F} \right) < -1$. This can arise only if we have (i) $\rho < 0$, and (ii) for sufficiently negative ρ , we also have $\sigma_T > \sigma_F$. To interpret this consider a setting where the total payoff $\tilde{v} = \tilde{v}_1 +$

$\tilde{v}_2 + \tilde{v}_C$, $\tilde{v}_F = \tilde{v}_1 + \gamma_F \tilde{v}_C$, $\tilde{v}_T = \tilde{v}_2 + \gamma_T \tilde{v}_C$, with $\gamma_F + \gamma_T = 1$ to preserve our original assumption about payoff structure, that $\tilde{v} = \tilde{v}_F + \tilde{v}_T$. Correlation arises because of the common component $\tilde{v}_C$, and will be negative when the coefficients γ_F and γ_T have different signs. For $\beta < 0$, besides having a significant common component between the firm and FIIs, it must also be the case that the FIIs' informational advantage σ_T must exceed the firm's informational advantage σ_F . A practical implication of this for empirical work is that if the estimate of the shallow parameter β is negative, that immediately tells us that FIIs must know more than firms.

3.4. Towards estimates of primitive parameters

Notice that in all we have the following primitive parameters in the model: σ_F , σ_T , σ_Z , and ρ . Of these σ_F can be estimated directly from the data on unexpected earnings, which proxies for v_F , and so v_F is observable. Given observability also of prices p and the aggregate FII order flow ω , it is clear that the parameters in the pricing rule shown in the table -- with β and λ defined in terms of primitive parameters -- can also be estimated almost as easily as coefficients β and λ in appropriate linear regressions.

Note that for meaningful estimates we need the following restrictions: $\rho \in (-1,1)$, $\sigma_i > 0, i = F, T, Z$. Another important empirical consideration is that we need to be careful when considering firm-windows where FII trades are zero. We need to distinguish between cases where these trades are zero because those firms do not have any FII ownership stakes, and cases where such stakes exist, so that the absence of any FII trade data should be interpreted as $\omega = 0$.

One way to view this empirical strategy is as a different response to multicollinearity. By construction, the variables v_F and the aggregate FII order flow ω are correlated: $\text{Cov}(v_F, \omega) = \tau_1 * \text{Var}(v_F) + \tau_2 * \text{Cov}(v_F, \omega)$. In the absence of a theory explaining the correlation, the most common traditional response would have been to regard this collinearity at best as a nuisance, and if it was deemed significant, to try and find instruments that are more orthogonal, to make it easier to isolate independent effects. With orthogonal regressors, each coefficient can be associated with a distinct factor, and interpretation becomes easier. Instead of treating collinearity as arising only from measurement problems, and as a nuisance to be minimized, we regard the collinearity as being related to fundamental information components, and explain it using an underlying theoretical model. So we treat that model not just as a source of qualitative predictions about regression coefficients, but as a direct bridge between what we can observe -- p, v_F, ω -- and primitive parameters -- σ_T, σ_Z , and ρ -- that we cannot observe. Therefore, we use the equilibrium model to quantify key primitive parameters of interest. There can be key economic information lurking in collinearity. A methodological contribution is to show how the underlying model of equilibrium allows us to learn more from the event study based on earnings announcements.

4. Empirical Analysis

4.1. Model

In our empirical analysis, we estimate the pricing function defined in Proposition 1. Specifically, we estimate a model that expresses returns at the earnings announcement date as a function of unexpected earnings, FII trading, and control variables:

$$ERET_{it} = \alpha + \beta UE_{it} + \gamma FIITR_{it} + \text{Control Variables} \quad (1)$$

where $\beta = 1 + \rho \left(\frac{\sigma_T}{\sigma_F} \right)$, $\lambda = \left(\frac{\sigma_T(\sqrt{1-\rho^2})}{\sigma_Z} \right)$, and

ERET = Earnings Announcement Return summed over the day of the earnings announcement and the following day, (0,1)

UE = Earnings per Share in quarter t less Earnings per Share from four quarters prior, scaled by share price two days before the earnings announcement

FIITR = Net Buying by all FIIs over the two-day earnings announcement window divided by shares outstanding on day -2.

Consistent with prior work on asset pricing, we include five factors that have been shown to explain the cross-section of returns (Fama and French (2015)). The factors are: market-wide return, firm size, book-to-market ratio, operating profitability scaled by assets, and prior asset growth. We explain our data sources and variable definitions in section 4.2.

What insights does estimation of Eq. (1) yield? To answer this, consider the pricing rule in Proposition 1: $p = \alpha + \beta v_F + \lambda \omega$, where

$$\alpha = 0, \quad \beta = 1 + \rho \left(\frac{\sigma_T}{\sigma_F} \right), \quad \lambda = \left(\frac{\sigma_T (\sqrt{1 - \rho^2})}{\sigma_Z} \right)$$

By mapping the regression model to this pricing rule, we can arrive at estimates of the primitive parameters. But there is a subtle econometric challenge involved. Notice that the three primitive variance parameters σ_F , σ_T , σ_Z , enter the equilibrium solution only in ratio form. These immediately implies that regardless of the estimation criterion we use (e.g. least squares, maximum-likelihood) we cannot from just this model identify all the three variance parameters. To see this, note that if a set of values for these three parameters optimizes any given criterion, then so will any scalar multiple of the same values. However, for σ_F we can construct an independent estimate from observed values of unexpected earnings (UE). We equate σ_F , the information advantage of the firm, to the sample estimate of the average

standard deviation of unexpected earnings (UE), across firms. Then in any implementation of the empirical model, we take that independently estimated value as a fixed value, and estimate the remaining three primitive parameters.

While our empirical model resembles a classical linear regression model, the linear coefficients of interest are themselves related nonlinearly to the primitive parameters. This creates a need to impose any estimation criterion (be it least-squares, maximum likelihood, or anything else) starting from first principles, rather than relying on canned software. To develop intuition, in the preliminary analysis we have completed so far, we also fixed one of the three remaining parameters (either σ_z or ρ) and estimated the remaining two. Given the definitions of β and λ in Proposition 1, we can invert them to obtain closed-form solutions for the remaining two parameters. For example, given ρ and σ_F and computed values of β and λ , it is easy to show that

$$\sigma_T = (\beta - 1) * \left(\frac{\sigma_F}{\rho}\right), \quad \sigma_Z = \left(\frac{\sigma_T}{\lambda}\right) * (\sqrt{1 - \rho^2})$$

And similarly, given σ_z and σ_F and computed values of β and λ , we can show that

$$\rho = \sqrt{A^2 / (1 - A^2)}, \text{ where } A = \left(\frac{(\beta-1)*\sigma_F}{\lambda*\sigma_Z}\right), \text{ and } \sigma_T = \left(\frac{(\beta-1)*\sigma_F}{\rho}\right)$$

For example, we vary ρ over the range (-0.9 to 0.9) in increments of 0.1, and for each of its values, estimate the two remaining parameters, σ_T and σ_z .⁴ Thus, our preliminary analysis provides estimates of the information advantage of FII traders (σ_T) and we are able to

⁴ We also calibrated σ_z at various positive values, starting from 0.001, and repeated this preliminary exercise. The following table summarizes what we found:

sigma_z	rho	sigma_t	sigma_f
0.001	non-est	0.0852	0.0814
0.01	0.6745	0.1157	0.0814
0.05	0.1798	0.434	0.0814
0.1	0.091	0.8575	0.0814
0.5	0.0183	4.2705	0.0814
1	0.0091	8.54	0.0814

For very small σ_z the estimates of intermediate quantities are not bounded. What this tells us is that for boundedness to hold σ_z cannot be too small, and for this sample, $\sigma_T \gg \sigma_F$, and $\rho > 0$.

compare the magnitude of this FII information advantage over its component, with the information advantage that firms have over the information that they reveal through earnings announcements (σ_F).

In ongoing empirical work we seek to estimate all three parameters independently. In that case we cannot get explicit closed-form solutions for the estimators of the primitive parameters. We also plan to estimate these primitive parameters at the level of each firm, to explore cross-sectional heterogeneity.

4.2. Data Sources and Variable Definitions

We obtain data from two sources: the PROWESS database of the Center for Monitoring Indian Economy Private Limited and the Securities and Exchange Board of India (SEBI) website. PROWESS provides daily firm returns, prices, and shares outstanding, financial statement data, and quarterly data on FII ownership. The quarterly financial information that we obtain consists of basic earnings per share (EPS) and the date of the board meeting on which financial results are approved. To measure control variables, we also obtain annual data on book value of common equity, operating profit, and total assets. While all Indian firms report parent-only unconsolidated financial statements, some of them simultaneously report consolidated financial statements. To maximize sample size, we examine only non-consolidated financial statement data. We collect quarterly ownership data to classify firms into those with FII ownership and those with zero FII ownership in the quarter surrounding the earnings announcement date.

Daily FII trading data is from the SEBI website. The database's basic unit of observation is trading activity by an FII for a stock on a trading day. In addition to an identifying code for each FII, the ISIN for the stock, and the exchange on which the

trades were executed, the database reports six measures of trading activity for each FII-stock-exchange-day quadruple: (a) the number of buys; (b) the number of sells; (c) aggregate shares bought on a day; (d) aggregate shares sold on a day; (e) value of shares bought; and (f) value of shares sold. Unfortunately, SEBI masks the FII identifying codes and changes the masks every month; consequently, FII-level analysis is difficult. Therefore, for each stock-trading day pair, we aggregate daily data across FIIs. Because we have no reason to expect exchange-related effects, we also aggregate daily trades across exchanges (primarily the BSE and the NSE). Thus, our unit of analysis is a stock-trading day pair. To compute FII trading at the earnings announcement date, we begin by defining FII net trading for firm i and day t as the difference between number of shares bought (BUY_{it}) and the number of shares sold ($SOLD_{it}$). We then aggregate this number over the two-day earnings announcement interval and deflate the sum by shares outstanding on day -2 to obtain FIITR.

We include the following control variables in the regression: market-wide return, firm size, book-to-market ratio, operating profitability scaled by lagged assets, and prior year asset growth. Market-wide return (MRET) is defined as the return on the CNX Nifty Index summed over days 0 and 1. The index daily return is calculated as the daily percentage change in the Index. Firm size is measured as the market capitalization on day -2 (MCAP). The book-to-market ratio (BTM) is obtained by dividing by the book value of equity at the end of the most recent fiscal year before the earnings announcement (year $t-1$) by MCAP on day -2. Operating Profitability is measured as profit before interest, tax, and depreciation for year $t-1$ divided by total assets at the end of year -2 (OPROF_A). Asset growth is the percentage change in total assets in year $t-1$ (AGRO).

Our sample period consists of thirteen years; it begins in the first quarter of 2003 (when SEBI began reporting data on FII trading at the stock level) and ends in the fourth quarter of 2015. Our initial sample consists of 138,689 firm-quarters for yy NSE and BSE

listed firms. We apply the following filters to arrive at our final sample. First, we exclude 47,715 observations that do not have data needed to compute unexpected earnings. Next, we exclude 23,975 observations that have missing data needed to compute control variables – previous year book value of equity, asset growth, or operating profitability. Our final sample consists of 66,999 firm-quarter observations.

Table 1 presents the sample distribution by year for three firm types: (a) FII-owned firms with non-zero trading during the earnings announcement date; (b) FII-owned firms that did not trade during the earnings announcement and (c) firms with no FII ownership. On average, 49% of the sample have no FII ownership and 32% have FII ownership but do not trade during earnings announcements. For about 19% of the sample, FIIs trade during the earnings announcement. The relative proportions of the three firm-types do not display a significant temporal shift during the sample period.

Table 2 reports univariate statistics for the variables in the regression model. The mean and median earnings announcement return are small and negative at -0.1% and -0.3%, respectively. Mean unexpected earnings scaled by share price is also negative at -0.6%; however, the median is slightly positive at 0.1%. The average net FII buying at the earnings announcement is zero, because the sample is dominated by non-trades. The FII net buying ranges from -0.5% to 0.5% as a percentage of shares outstanding. Turning to the control variables, mean market return is positive at 0.3%, the mean market capitalization is about ₹17 Billion, and the mean book-to-market ratio is 1.1. The sample firms are profitable on average and are growing – mean operating profitability is 14.6% and mean asset growth is 18.2%

Table 3 reports estimates of the primitive parameters from a regression based on one assumption about ρ . Examining all of the estimates from the entire calibration-cum-

estimation exercise in this preliminary analysis⁵, one conclusion is striking. The estimate for σ_T is always significantly larger than for σ_F . Also when we fix σ_Z and estimate ρ along with σ_T , the estimate for ρ is always significantly different from zero. So allowing for correlation matters. These significant conclusions in this event study are possible only because we explicitly model the underlying equilibrium in a correlated environment.⁶

5. Conclusion

How informed are Foreign Institutional Investors (FIIs) in Indian financial markets? We build a model of imperfect competition in a financial market, in which market makers have up to two public signals on which to define their pricing rule – firms’ earnings announcements and FII trading signals. We define the payoff structure with a sum of correlated components, one known to firms, and the other, to FIIs. We then estimate the deep parameters of the model such as the variance governing the FIIs’ informational advantage, the level of background noise, and the correlation between the two components of the payoff. These are parameters of interest but cannot be directly observed, or inferred from the shallow parameters of an econometric model. The underlying model of equilibrium serves as a bridge between what we can observe and what we are also interested in but cannot. So instead of treating multicollinearity as a problem to be resolved using more orthogonal instruments, or a better selection of regressors, we harness it by explaining it using an underlying model of financial market equilibrium, and identify deep parameters that are unobservable but have economic significance.

⁵ Also see footnote 4 above.

⁶ We also examined measures of informativeness. In our context we can define various measures of ex ante and ex post measures. For example, an example of an ex ante measure of informativeness is $\frac{\sigma_T}{\sigma_F}$, which measures the information advantage of FIIs at the time of an earnings announcement, relative to the information advantage of firms about what is released in the earnings announcement. For ex post measures, we look at the total variance of abnormal returns (adjusted for control variables), and ask what percentage of this variance is explained by unexpected earnings or FII trading? We find that both unexpected earnings and FII trading each explain slightly less than 1% of the total variation at the time of earnings announcement.

Our results indicate the information advantage of FIIs with respect to the component they have information about, exceeds the information advantage that firms have with respect to information released via earnings announcements. We also note that correlation between the two fundamental information components is significant, so that in econometric work we should consider such an environment, even beyond any correlation arising from imperfect measurement.

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Appendix

Given our assumption about component payoff structure, the multinormal random vector $\tilde{y} \equiv$

$\text{tr}\{\tilde{v}_F, \tilde{v}_T, \tilde{z}\}$, where “tr” denotes the transpose, is

$$\begin{bmatrix} \tilde{v}_F \\ \tilde{v}_T \\ \tilde{z} \end{bmatrix} \sim MN \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_F^2 & \rho \cdot \sigma_F \cdot \sigma_T & 0 \\ \rho \cdot \sigma_F \cdot \sigma_T & \sigma_T^2 & 0 \\ 0 & 0 & \sigma_z^2 \end{bmatrix} \right\}. \text{ Let us call this 3x3 variance-covariance}$$

matrix Σ , and let $\text{tr}(j) \equiv \{1, 1\}$. Let $\tilde{y}_1 \equiv \text{tr}\{\tilde{v}_F, \tilde{v}_T\}$, and Σ_{11} be the leading 2x2 minor of

Σ . Then total payoff $\tilde{v} \equiv \text{tr}(j) \cdot \tilde{y}_1$, and $\tilde{v} \sim N(0, \text{tr}(j) \cdot \Sigma_{11} \cdot j)$.

We now define the FII trader’s optimization problem. Since the trader can observe

announced earnings v_F , her own private information v_T , and FII noise trade z , and faces

the pricing rule $p = \alpha + \beta v_F + \lambda \omega$, where aggregate FII order flow $\omega = x + z$, the

problem of the trader is to choose a demand x to maximize profit, defined by $((v_F + v_T) -$

$(\alpha + \beta v_F + \lambda(x + z)))x$ which yields the first-order condition $-\alpha + (1 - \beta)v_F + v_T +$

$\lambda z = 2\lambda x$. Solving for x yields $\tau_0 = \frac{-\alpha}{2\lambda}$, $\tau_1 = \left(\frac{(1-\beta)}{2\lambda}\right)$, $\tau_2 = \left(\frac{1}{2\lambda}\right)$, $\tau_3 = -\left(\frac{1}{2}\right)$.

Then aggregate order flow $\omega = x + z = \frac{-\alpha}{2\lambda} + \tau_1 v_F + \tau_2 v_T + \left(\frac{1}{2}\right)z$.

We then compute the expectation $E(\tilde{v}|v_F, \omega)$ where $\tilde{v} = \tilde{v}_F + \tilde{v}_T$. Define the multinormal

random vector $\tilde{h} \equiv \text{tr}\{\tilde{v}, \tilde{v}_F, \tilde{\omega}\}$, where “tr” denotes the transpose.

$$\begin{bmatrix} \tilde{v} \\ \tilde{v}_F \\ \tilde{\omega} \end{bmatrix} \sim MN \left\{ \begin{bmatrix} 0 \\ 0 \\ \frac{-\alpha}{2\lambda} \end{bmatrix}, \begin{bmatrix} \text{tr}(j) \cdot \Sigma_{11} \cdot j & \sigma_F^2 + \rho \cdot \sigma_F \cdot \sigma_T & 0 \\ \sigma_F^2 + \rho \cdot \sigma_F \cdot \sigma_T & \sigma_F^2 & 0 \\ 0 & 0 & \sigma_z^2 \end{bmatrix} \right\}.$$

Because of multinormality, the expectation $E(\tilde{v}|v_F, \omega)$ is linear in the conditioning

arguments. Recall that by virtue of market efficiency we have $p = E(v|v_F, \omega)$. Therefore,

we equate corresponding coefficients to get three equations of the form, $\alpha = f_1(\alpha, \beta, \lambda)$, $\beta =$

$f_2(\alpha, \beta, \lambda)$, $\lambda = f_3(\alpha, \beta, \lambda)$. From the first alone, it is easy to show that $\alpha = 0$. Manipulating

the other two leads to a cubic in two variables, β and λ , instead of in λ alone as in Kyle

(1985) and Rochet and Vila (1994). We obtain three candidate solutions of which only one

satisfies $\lambda > 0$, which is needed to satisfy second-order conditions. So we have a unique real root. The solution is easily verified. Plugging the equilibrium values of β and λ into the trader's strategy coefficients yields Proposition 1.

Table 1**Yearly Distribution of FII Trading during Earnings Announcements**

This Table reports the sample distribution by year for three types of firm-quarters: (a) firm-quarters with FII trading during earnings announcements, firm-quarters with no FII trading during earnings announcements, and firm-quarters with zero FII ownership. The sample period is from March 2003 to December 2015. Data on FII trading are obtained from the SEBI website: <http://www.sebi.gov.in>. Earnings announcement dates and earnings per share, stock prices and returns, and quarterly FII ownership levels are obtained from the PROWESS database.

Year	FII Trading during earnings announcement		No FII Trading around earnings announcement		No FII Ownership		Total
	# obs.	%	# obs.	%	# obs.	%	# obs.
2004	396	14%	1,007	35%	1,510	52%	2,913
2005	893	18%	1,633	33%	2,419	49%	4,945
2006	1,041	20%	1,704	32%	2,587	49%	5,332
2007	1,206	21%	1,747	31%	2,695	48%	5,648
2008	1,135	20%	1,919	34%	2,651	46%	5,705
2009	984	17%	1,895	34%	2,766	49%	5,645
2010	1,143	20%	1,742	30%	2,933	50%	5,818
2011	1,167	19%	1,785	30%	3,052	51%	6,004
2012	1,247	19%	2,055	31%	3,325	50%	6,627
2013	1,256	20%	2,109	33%	2,963	47%	6,328
2014	1,352	19%	2,293	32%	3,557	49%	7,202
2015	1,004	21%	1,453	30%	2,375	49%	4,832
Total	12,824	19%	21,342	32%	32,833	49%	66,999

Table 2
Descriptive Statistics

This table presents descriptive statistics for the variables used in our analysis. ERET is the earnings announcement return obtained by summing returns over days (-1, +1) relative to the earnings announcement date. UE equals earnings per share for quarter t less earnings per share for quarter t-4 divided by closing price on day -2 relative to the earnings announcement for quarter t-1. FIITR is net FII buying over the earnings announcement period, days (0, +1), divided by shares outstanding on day -2. Net FII buying for a firm on a day equals number of shares bought less number of shares sold for that firm by all FIIs on that day. Market-wide return (MRET) is defined as the return on the CNX Nifty Index summed over days 0 and 1. The index daily return is calculated as the daily percentage change in the Index. Firm size (MCAP) is measured as the market capitalization on day -2. The book-to-market ratio (BTM) is obtained by dividing by the book value of equity at the end of the most recent fiscal year before the earnings announcement (year t-1) by MCAP on day -2. Operating Profitability (OPROF_A) is measured as profit before interest, tax, and depreciation for year t-1 divided by total assets at the end of year -2. Asset growth (AGRO) is the percentage change in total assets in year t-1. The sample consists of 66,999 earnings announcements from March 2003 to December 2015. Data on FII trading are obtained from the SEBI website: <http://www.sebi.gov.in>, and earnings announcement dates, earnings per share, shares outstanding, firm stock prices, returns, market capitalization, book value of equity, profit before interest, tax, and depreciation, and total assets are obtained from the PROWESS database.

	# obs.	Mean	Median	Std. Dev.	Min	Max
ERET	66,999	-0.1%	-0.3%	6.3%	-34.5%	39.9%
UE	66,999	-0.6%	0.1%	13.7%	-135.3%	132.8%
FIITR	66,999	0.0%	0.0%	0.1%	-0.5%	0.5%
MRET	66,999	0.3%	0.3%	2.6%	-16.2%	12.5%
MCAP (₹Million)	66,999	16,978.3	738.2	68,757.2	8.6	844,499.3
BM	66,999	1.1	0.8	2.1	-16.0	17.5
OPROF_A	66,999	14.6%	12.6%	13.1%	-14.0%	96.9%
AGRO	66,999	18.2%	9.5%	40.9%	-48.0%	446.5%

Table 3
Estimates of Primitive Parameters

The Table reports estimates of primitive parameters from estimating the following model:

$$ERET_{it} = \alpha + \beta UE_{it} + \gamma FIITR_{it} + \text{Control Variables} \quad (1)$$

Primitive parameters are obtained by mapping the estimates from the above model to the following equation:

$$\alpha = 0, \beta = 1 + \rho \left(\frac{\sigma_T}{\sigma_F} \right), \lambda = \left(\frac{\sigma_T(\sqrt{1 - \rho^2})}{\sigma_Z} \right)$$

The four primitive parameters are σ_T , σ_Z , ρ , and σ_F . Of the four, we equate σ_F , the information advantage of the firm, to the sample estimate of the average standard deviation of unexpected earnings (UE), across firms. Of the remaining three parameters, we vary ρ over the range (-0.9 to 0.9) in increments of 0.1, and for each of its values estimate the two remaining parameters, σ_T and σ_Z . The Table provides estimates for the case of $\rho = -0.9$. Variable Definitions and Data Sources are described in Table 2.

Variable	Coefficient	t-statistic	p-value
σ_T	0.0975	556.96	0.00
σ_Z	0.0146	19.36	0.00
MRET	0.6124	68.49	0.00
MCAP	0.0000	-0.72	0.47
BM	-0.0003	-3.16	0.00
OPROF_A	-0.0038	-2.49	0.01
AGRO	0.0011	1.73	0.08
Adjusted R ²			7.81%
Obs.			66,999